Verification of Conjecture and Enumeration Tables

Summary of `verify_conjecture_and_tables.py`

(a) Theory

Tooth generating functions

The paper defines generating functions $h_r(q)$ for teeth of base length r, together with normalized generating functions $u_r(q)$ through

$$h_r(q) = \frac{q^r}{(q;q)_r} u_r(q).$$

The functions h_r satisfy the two-term recurrence

$$(1-q^r)h_r(q) = 2q\,h_{r-1}(q) - q^2 h_{r-2}(q),$$

while the normalized functions satisfy

$$u_r(q) = 2u_{r-1}(q) - (1-q^{r-1})u_{r-2}(q),$$

with $u_0 = u_1 = 1$.

Conjecture 3

The paper conjectures a Durfee-type decomposition for the bivariate generating function

$$U(t) = \sum_{r\geq 0} u_r(q) t^r,$$

namely

$$U(t) = \sum_{m\geq 0} \frac{q^{m^2} t^{2m}}{(t;q)_m (t;q)_{m+1}}.$$

The quantity m plays the role of a Durfee-square size, analogous to the classical Durfee decomposition of integer partitions.

Coefficient triangle

The coefficients

$$T(r,N) = [q^N]h_r(q)$$

count teeth of base length r containing N cubes. These coefficients form the triangular array presented in Table 2 of the paper.

(b) What the script does

The script performs two independent verification tasks.

Part 1: Conjecture verification

It computes the generating function $U(t)$ in two different ways:

1. From the recurrence for $u_r(q)$;
2. From the conjectured Durfee-type series.

The script then expands both expressions as power series in t and checks that the coefficients of $t^0, \ldots, t^{15}$ agree exactly.

Part 2: Table verification

It computes the coefficient triangle

$$T(r,N) = [q^N]h_r(q)$$

for

- $r = 1, \ldots, 7$;
- $N = 1, \ldots, 15$;

and reproduces the values reported in Table 2 of the paper.

It additionally:

- computes the column sums $\sum_{r=1}^{N} T(r,N)$;
- verifies the closed-form identity

$$T(2,N) = N - 1.$$

(c) How the script does it

Construction of $h_r(q)$

`compute_h(max_r)` generates the functions $h_r(q)$ recursively using

$$h_r(q) = \frac{2q\,h_{r-1}(q) - q^2 h_{r-2}(q)}{1 - q^r}.$$

The resulting expressions are rational functions in q, simplified using SymPy's `cancel`. The script calls `compute_h(8)` rather than `compute_h(7)`: the extra row is a buffer used internally by the series-expansion arithmetic and is not printed.

Construction of $u_r(q)$

`compute_u(max_r)` generates the normalized functions using

$$u_r(q) = 2u_{r-1}(q) - (1 - q^{r-1})u_{r-2}(q).$$

Because the recurrence has polynomial coefficients and polynomial initial values, every $u_r(q)$ is itself a polynomial. The implementation therefore uses `sp.expand` rather than `sp.cancel`; no denominator is ever produced. The function is called with argument $N_{\max} + 2 = 17$ rather than $N_{\max} = 15$; the extra two terms act as a buffer for the truncated series comparison and are not themselves reported.

q-Pochhammer symbols

`poch(a,n)` evaluates the finite q-Pochhammer product

$$(a;q)_n = \prod_{k=0}^{n-1}(1 - aq^k).$$

These factors appear in the denominator of the conjectured Durfee expansion.

Verification of the conjecture

The script constructs

$$U_{\text{recurrence}}(t) = \sum_{r=0}^{N_{\max}} u_r(q)t^r$$

from the recurrence and independently constructs

$$U_{\text{conjecture}}(t) = \sum_{m\geq 0} \frac{q^{m^2}t^{2m}}{(t;q)_m(t;q)_{m+1}}.$$

Each term of the conjectural series is expanded in powers of t, truncated at order t^{15}, and accumulated. The sum over m runs from 0 to $\lfloor N_{\max}/2 \rfloor + 2 = 9$: since the m-th term begins at order t^{2m}, terms with $m > N_{\max}/2$ contribute nothing below $t^{N_{\max}}$, so this truncation is exact rather than approximate.

The coefficients of t^r are then compared symbolically for every

$$0 \leq r \leq 15.$$

Any mismatch would be reported explicitly.

Computation of the coefficient triangle

For each $h_r(q)$, the script expands the power series in q up to order q^{15}.

The coefficient

$$T(r, N)$$

is extracted directly using SymPy's coefficient routines and stored in a lookup table.

Auxiliary checks

Two additional consistency checks are performed:

1. Column sums:

$$\sum_{r=1}^{N} T(r, N),$$

which count all teeth containing exactly N cubes. These are the values of OEIS sequence A001523 (number of weakly unimodal compositions of N).

2. The row-2 identity:

$$T(2, N) = N - 1.$$

(d) Output produced

Conjecture verification

The script reports whether every coefficient through t^{15} matches between the recurrence-generated and conjectured forms of $U(t)$.

A successful run prints

```
All coefficients match (r = 0 to 15): True
```

providing symbolic evidence for the conjecture up to the tested order.

The script also prints the first several polynomials

$$u_0(q), u_1(q), \ldots, u_7(q),$$

which allows inspection of the normalized generating functions appearing in the conjecture.

Coefficient triangle

The complete table

$$T(r, N), \qquad 1 \le r \le 7,\ 1 \le N \le 15,$$

is printed in matrix form and can be compared directly with Table 2 of the paper.

Column sums

The values

$$\sum_{r=1}^{N} T(r, N)$$

for $N = 1, \ldots, 9$ are displayed. These values form OEIS sequence A001523 (number of weakly unimodal compositions of N), providing an independent enumeration sequence and connecting the output directly to the paper's Sequences section.

Row-2 verification

The script prints whether

$$T(2, N) = N - 1$$

holds throughout the computed range.

(e) What we learn

The conjectured Durfee-type expansion survives a nontrivial symbolic test. The recurrence definition of $u_r(q)$ and the conjectured closed-form series produce identical coefficients through order t^{15}, providing strong computational evidence for Conjecture 3.

The recurrence and the conjecture encode the same structure. The agreement demonstrates that the Durfee-style decomposition correctly reproduces all normalized generating functions $u_r(q)$ within the tested range.

Table 2 is independently reproducible. The coefficient triangle is generated directly from the recurrence for $h_r(q)$, confirming that the tabulated values follow from the theory rather than manual calculation.

Several internal consistency checks hold automatically. The column sums and the identity $T(2, N) = N - 1$ emerge naturally from the computed coefficients, providing additional validation of the generating-function framework.

The script serves as a verification tool rather than a discovery tool. Its primary purpose is to test the mathematical results of the paper symbolically and to reproduce the published tables exactly using independent calculations.